

07 Logistic Regression and Gradient Descent

Markéta Tranberg, mlap@viauc.dk

March 2026

Table of Contents

1 A bit of calculus

2 Gradient descend

3 Logistic regression

4 Code example

Derivative at a point

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

- The slope of the tangent line to the curve of $f(x)$ at $x = x_0$
- The rate of change of $f(x)$ with respect to x

Finding extrema of a function

If f has a local maxima or **minima** at an interior point c of its domain and if $\frac{df}{dx}$ is defined at c , then

$$f'(c) = 0$$

Derivatives of common functions

- **Constant:** $\frac{d}{dx}(c) = 0$
- **Polynomial:** $\frac{d}{dx}(x^n) = nx^{n-1}$
- **Exponential:** $\frac{d}{dx}(e^x) = e^x$ and $\frac{d}{dx}(a^x) = a^x \ln a$
- **Logarithmic:** $\frac{d}{dx}(\ln x) = \frac{1}{x}$ and $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$
- **Trigonometric:** $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$

Gradient vector

$$\nabla f(x_1, x_2, \dots, x_n) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

where $\frac{\partial f}{\partial x_1}$ is a partial derivative of f wrt. x_1 .

Example: gradient vector

$$f(x) = x^2$$

$$\nabla f =$$

$$f(x, y) = 2x + y^2$$

$$\nabla f =$$

Table of Contents

- 1 A bit of calculus
- 2 Gradient descend**
- 3 Logistic regression
- 4 Code example

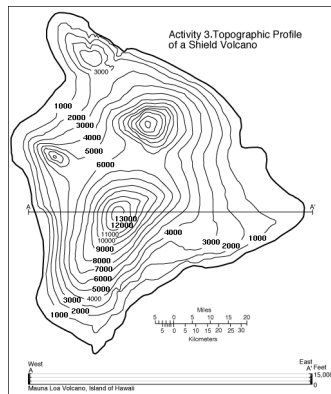
Topographic map analogy

Close lines = steep terrain

Far apart lines = flat terrain

Gradient ∇f = vector pointing in direction of the steepest ascent

Learning rate = How far you walk before checking map again



Gradient descent

What: finding minima of a function

How: finding direction of steepest slope and taking a step in opposite direction

How quickly: Speed is dependent of step size γ (learning rate)

$$x_{t+1} = x_t - \gamma_t \nabla f(x_t)$$

Note: at the minimum, the slope of tangent (derivative) is 0.

Learning rate

Learning rate $\gamma = \text{step size}$

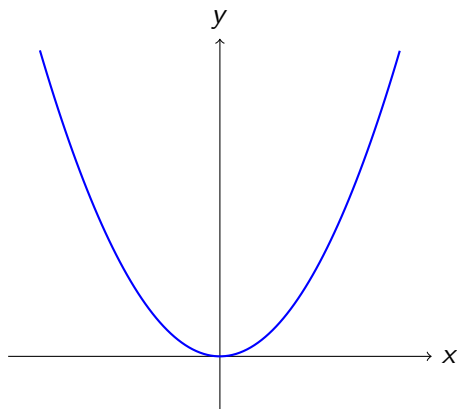


Table of Contents

- 1 A bit of calculus
- 2 Gradient descend
- 3 Logistic regression**
- 4 Code example

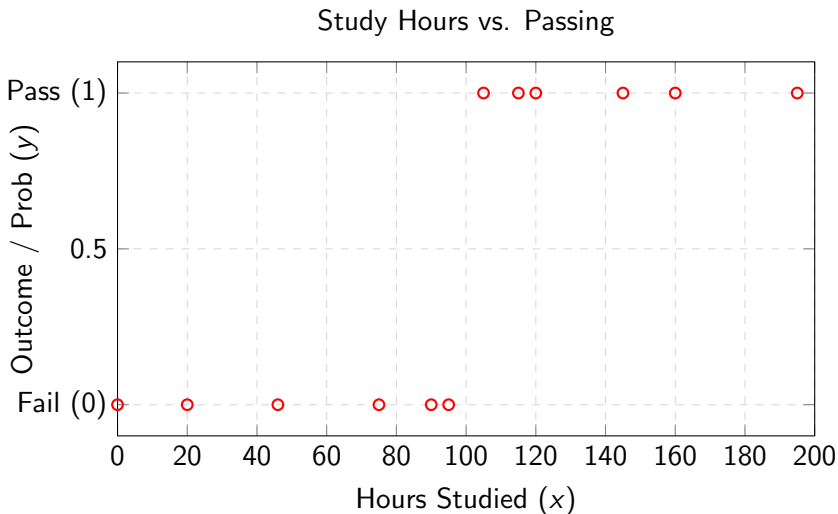
Logistic regression

Classification algorithm that returns a continuous probability
(that's why regression)

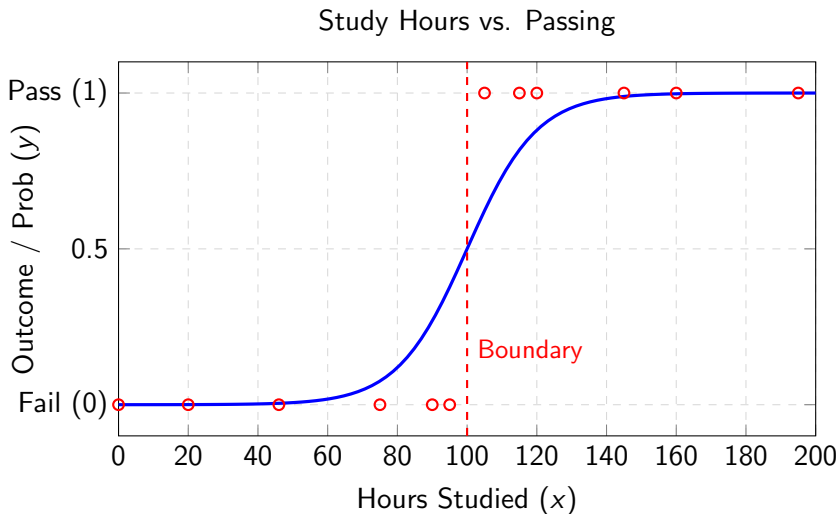
$$z = wx + b$$

- z : "logit", output deciding a class
- w : weight, importance of the feature(s)
- x : input, features
- b : bias, offset

Visualizing Logistic Regression: Exam Success



Visualizing Logistic Regression: Exam Success



Sigmoid function

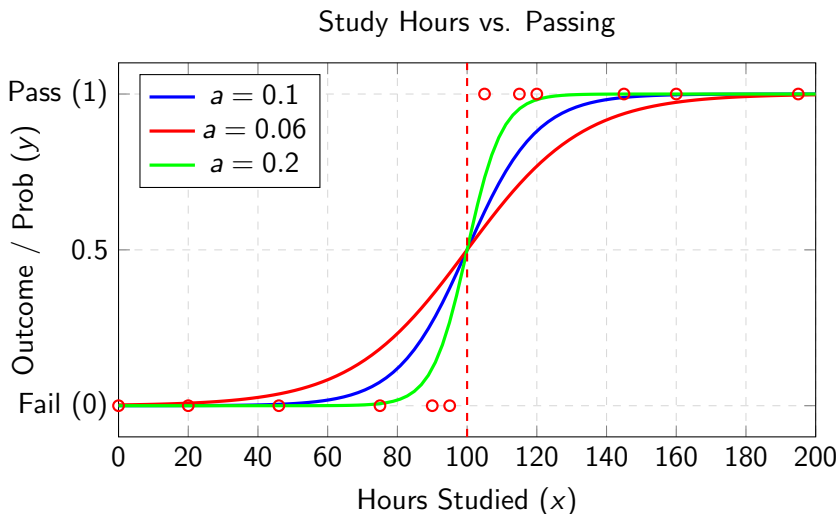
Sigmoid function maps any real number into a range between 0 and 1 - converts any output to a probability.

$$p(X) = \frac{1}{1 + e^{-(ax+b)}}$$

Note: $z = 0$, $p(X) = 0.5$. As z grows larger, $p(X)$ approaches 1. As z grows smaller (large negative), $p(X)$ approaches 0.

Sigmoid function translates z into **confidence**.

Impact of coefficient on sigmoid function



Connection to gradient descent

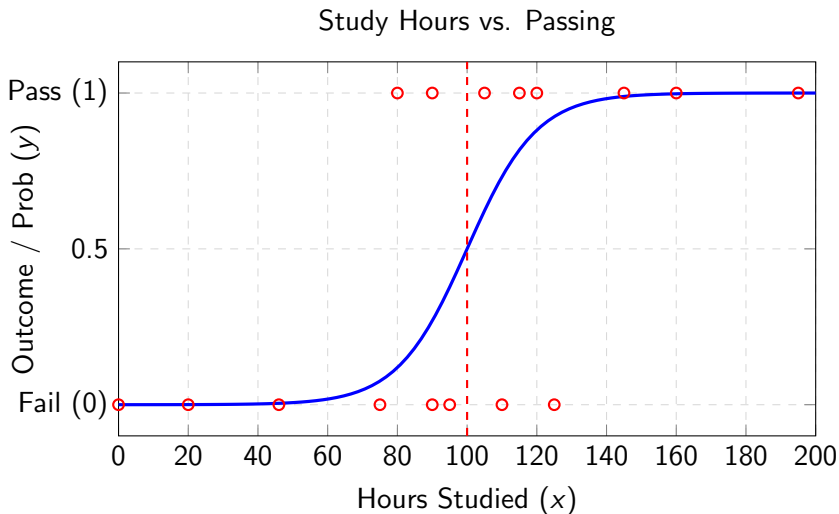
Log loss function: convex parabola (to prevent gradient descent from getting stuck)

Log loss function penalizes confident wrong predictions (true 0, predict 0.99)

At minima of loss function: best coefficients w and b ($z = wx + b$)

Learning = looking for the "best" split = minimal loss function

Linearly non-separable data



Multiple features

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

Higher weight value w_i = higher impact on z (weighted voting system)

Positive w_i = vote for

Negative w_i = vote against

Table of Contents

- 1 A bit of calculus
- 2 Gradient descend
- 3 Logistic regression
- 4 Code example**

Code

Github link: https://github.com/304125/MAL1_S26/tree/master/07_Logistic_Regression_and_Gradient_Descent